INCH-POUND

MIL-S-19500/201B
25 September 1989
SUPERSEDING
MIL-S-19500/201A
26 April 1962

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, GERMANIUM TYPE 1N277 JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a germanium semiconductor diode. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

V_R	V_{RM}	I_O 1/	I_F	$I_{F(surge)}$
V(pk)	V(pk)	mA dc	mA	1/120 s
125	100	50	270	a 1

1/ Derate 0.77 mA dc/°C above T_A = +25°C.
Operating ambient temperature: -65°C to +75°C.
Storage temperature: -65°C to +100°C.
Barometric pressure, reduced: 8 mmHg.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ES, 1507 Wilmington Pike, Dayton, OH 45444-5276, by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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1.4 Primary electrical characteristics. The diode shall have salient characteristics at ambient temperature of +25°C ±3°C, unless otherwise specified.

	V_F	V_{R1}	I_{R2}	I_{R3}	I_{R4}	I_R	I_R	V_F
	I_F = 5 mA dc	V_R = 10 V dc	V_R = 50 V dc	V_R = 100 V dc	V_R = 125 V pk	V = 10 V dc	V = 50 V dc	I_F = 100 mA dc
					AC method	T_A = +75°C	T_A = +75°C	
Min	0.3 V dc							
Max	0.5 V dc	10 µA dc	50 µA dc	400 µA dc	1.0 mA (pk)	75 µA dc	25 µA dc	1.0 V dc

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

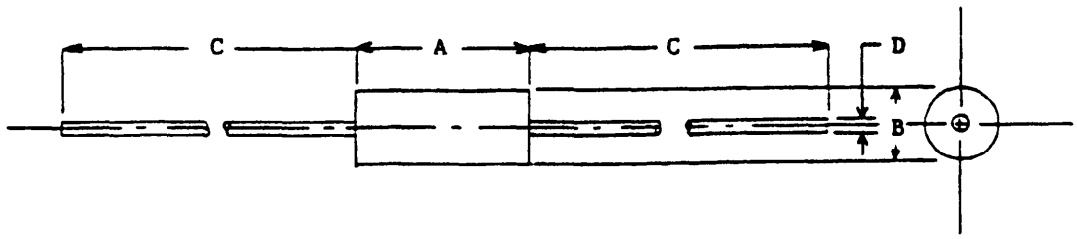
(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.230	.300	5.84	7.62	
B	.085	.130	2.16	3.30	4
C	1.000	1.500	25.40	38.10	3
D	.018	.022	0.46	0.56	1, 2, 3

NOTES:
1. The specified lead diameter applies in the zone between .050 (1.27 mm) and 1.000 (25.40 mm) from the diode body. Outside of this zone, the lead diameter is not controlled.
2. Gold plated, tinned, or solder plated leads may be supplied providing units conform to subgroups 2 and 4 of group B inspection and subgroup 1 of group C inspection.
3. Both leads shall be within the specified dimension.
4. The minimum body diameter shall be maintained over .15 (0.38 mm) of body length.
5. Metric equivalents are given for general information only.

FIGURE 1. Semiconductor device, diode, type 1N277 (DO-7).

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3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figure 1.

3.3.1 Lead material and finish. Lead finish shall be Dumet wire. Lead finish shall be tin plated or solder plated. Where a choice of lead material or finish is desired, it shall be specified in the contract or purchase order (see 6.3).

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, the following marking may be omitted from the body of the diode.

a. Manufacturer's identification.

b. Country of origin.

c. The "1N" portion of the type designation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-STD-19500)	Measurement
	JANTX and JANTXV levels
3	T_A = maximum rated storage temperature
10	Test condition C, T_A at 55°C, t = 48 hours minimum
11	I_{R1} and V_{F1}
12	See 4.3.1
13	Subgroup 2 of table I herein; ΔI_{R1} = 100 percent of initial reading or 5 µA whichever is greater. ΔV_{F1} = ±10 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: All devices shall be operated at +25°C ±3°C under one of the following conditions:

Condition 1: See 4.5.3
I_O = 50 mA dc
V_R = 100 V pk

Condition 2: I_F = 50 mA dc

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table IV herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb of MIL-S-19500 and table IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Time limit for end points. End points for qualification and quality conformance inspection shall be completed within 96 hours after completion of the last test in the subgroup.

4.5.3 Power burn-in criteria. This test shall be conducted with a half-sine wave of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be not greater than 180° nor less than 130°, and the power shall be equal to or greater than that of a half-sine waveform.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Lead material and finish (see 3.3.1).

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Forward voltage	4011	I_F = 5 mA dc (see 4.5.1)	V_{F1}	0.3	0.5	V dc
Forward voltage	4011	I_F = 100 mA dc (see 4.5.1)	V_{F2}		1.0	V dc
Reverse current	4016	V_R = 10 V dc	I_{R1}		10	µA dc
Reverse current	4016	V_R = 50 V dc	I_{R2}		50	µA dc
Reverse current	4016	V_R = 100 V dc	I_{R3}		400	µA dc
Peak reverse current at peak reverse voltage (ac method)	4016	V_R = 125 V(pk)	i_r		1.0	mA(pk)
Subgroup 3		T_A = +75°C				
Reverse current	4016	V_R = 10 V dc	I_{R4}		75	µA dc
Reverse current	4016	V_R = 50 V dc	I_{R5}		250	µA dc
Subgroups 4 and 5						
Not applicable						
Subgroup 6						
Surge current	4066	I_O = 50 mA dc; $i_{f(surge)}$ = 1.0; t_p = 17120 s; 10 surges at 1 per minute				
Electrical measurements		See table IV, steps 1, 2, 4, and 6				

1/ For sampling plan, see MIL-S-19500.

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TABLE IIb. Group B inspection for JAN, JANTX, and JANTXV.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Solderability	2026	
Resistance to solvents	1022	
Subgroup 2		
Thermal shock (temperature cycling)	1051	Test condition A
Hermetic seal	1071	
a. Fine leak		Test condition H
b. Gross leak		Test condition E
Electrical measurements		See table IV, steps 1, 2, 4, and 6
Subgroup 3		
Steady-state operation life	1027	f = 60 Hz; I_O = 50 mA dc; V_R = 100 V(pk); t = 340 hours; T_A = +25°C; see 4.5.3
Electrical measurements		See table IV, steps 1, 3, and 5
Subgroup 4		
Decap internal visual (design verification)	2075	Visual criteria in accordance with qualified design.
Subgroup 5		Qualifications only
Thermal resistance	4081	
Subgroup 6		
High temperature life (nonoperating)	1032	T_A = +100°C; t = 340 hours
Electrical measurements		See table IV, steps 1, 3 and 5

1/ For sampling plan, see MIL-S-19500.

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TABLE III. Group C inspection (all quality levels).

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Physical dimensions	2066	
Subgroup 2		
Thermal shock (glass strain)	1056	Test condition A
Terminal strength	2036	Test condition A; 4 pounds; t = 15 ±3 s
Hermetic seal	1071	
a. Fine leak		Test condition H
b. Gross leak		Test condition E
Moisture resistance	1021	
External visual	2071	
Electrical measurements		See table IV, steps 1, 2, 4, and 6
Subgroup 3		
Shock	2016	Nonoperating at 1,500 g's; t = 0.5 ms; 5 blows each orientations X, Y, and Y_2
Vibration, variable frequency	2056	
Constant acceleration	2006	Nonoperating at g's = 20,000 minimum, 1 minute in each orientation X_1, Y_1, and Y_2
Electrical measurements		See table IV, steps 1, 2, 4, and 6
Subgroup 4		
Salt atmosphere (corrosion)	1041	
Subgroup 5		
Not applicable		
Subgroup 6		
Steady-state operation life	1026	f = 60 Hz; T_A at +25°C; I_O = 50 mA dc; V_R at 100 V pk; t = 1,000 hours; see 4.5.3
Electrical measurements		See table IV, steps 1, 3, and 5

1/ For sampling plan, see MIL-S-19500.

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TABLE IV. Groups A, B, and C electrical end-point measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	I_F = 100 mA dc	V_{F2}		1.0	V dc
2.	Reverse current	4016	V_R = 10 V dc	I_{R2}		10	µA dc
3.	Reverse current	4016	V_R = 10 V dc	I_{R6}		15	µA dc
4.	Reverse current	4016	V_R = 50 V dc	I_{R4}		50	µA dc
5.	Reverse current	4016	V_R = 50 V dc	I_{R7}		75	µA dc
4.	Peak reverse current at peak reverse voltage	4016	V_R = 125 V(pk)	i_r		1.0	mA(pk)

6.3 Substitution information. Devices covered by this specification are substitutable for the manufacturers' and users' Part or Identifying Number (PIN). This information in no way implies that manufacturers' PIN's are suitable as a substitute for the military PIN.

Military PIN	Manufacturers' CAGE code	Manufacturers' and users' PIN

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR
Navy - SH
Air Force - 11, 85

User activities:
Army - MI, SM
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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